Monetary Policy Announcements, Asset Prices, and Portfolio Reallocation

A Research Proposal

Outline

1. Background & Rationale
2. Research Aim & Objectives
3. Literature Review & Hypotheses
4. Methodology
5. Expected Findings & Contributions
6. Preliminary Evidence (Corrected Data, Oct 2026)
7. Research Plan
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Background & Rationale

The 2022–2025 Tightening Cycle: A Stress Test

Unprecedented in Four Decades

- Fed funds rate: 0.25% $\rightarrow$ 5.50% in 18 months
- Simultaneous selloffs in equities and bonds
- 60/40 portfolio posted -17% in 2022
- Negative stock-bond correlation broke down

Key Question

Does FOMC communication contain **incremental information** beyond the rate decision itself, that drives asset price movements?

Three Gaps in the Literature

Methodological Gap

Kuttner (2001) → JK (2020) → Bauer-Swanson (2023): identification evolved, but rarely integrated with NLP-based sentiment analysis.

Empirical Gap

2022–2025 cycle lacks systematic study: rapid hikes, QT, inflation shocks, and shifted communication regime are poorly understood.

Measurement Gap

Dictionary-based sentiment (LM, CB) is coarse; FinBERT is available but lack of large-scale reproducible pipeline for FOMC statements.

This proposal addresses all three gaps simultaneously.

Research Aim & Objectives

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To quantify the **incremental information content** of FOMC communication beyond policy rate surprises, and to examine its **differential transmission** across asset classes and **regime-dependent** effects on portfolio reallocation.

Contributing to: monetary economics · central bank communication · asset pricing

Four Objectives & Four Hypotheses

	Objective	Hypothesis
O1	Measure sentiment-surprise overlap	H_1: $0 < \beta < 1$
O2	Test incremental explanatory power	H_2: $\beta_2 \neq 0$
O3	Map sentiment to structural shocks	H_3: $\lvert\gamma_2\rvert > \lvert\gamma_1\rvert$
O4	Identify regime-dependent effects	H_4: $\beta_3 \neq 0$

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- **O2:** Does sentiment explain asset returns *after* controlling for rates?

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- **O3:** Does sentiment capture the *information* shock, not just policy shock?

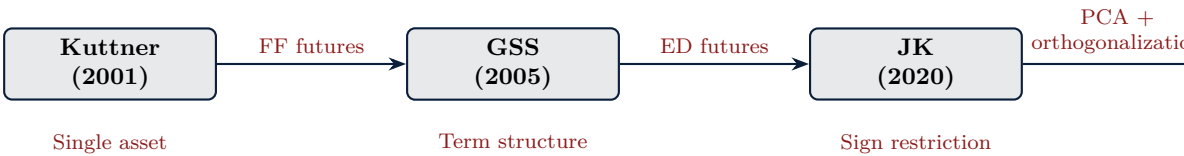
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- **O4:** Are effects stronger during forward guidance / ZLB periods?

Literature Review & Hypotheses

Monetary Policy Surprise: Identification Evolution



- **Key insight:** identification moved from *single-asset* to *multi-asset PCA*, addressing the "information effect" (Nakamura & Steinsson, 2018)
- Gap: None of these integrate **textual sentiment** as a separate shock

Central Bank Communication: NLP Evolution

Dictionary-Based

- Loughran & McDonald (2011): finance-specific word lists
- Apel & Blix (2014): CB hawkish/dovish dictionary
- Limitation: no context, no negation, no sarcasm

ML-Based

- FinBERT (Araci, 2019): fine-tuned BERT
- Ahrens (2025): multimodal transformer
- Advantage: context-aware, scalable
- Gap: lack of reproducible FOMC pipeline

This proposal: hybrid approach —LM+CB dictionary → FinBERT validation → cross-robustness

Hypothesis 1: Sentiment ∼ Surprise

H_1: Sentiment is correlated with but not redundant to surprise

$$\text{Sent}_t = \alpha + \beta\, S_t + \varepsilon_t \tag{1}$$

Test: $0 < \beta < 1$ (significant but $\beta \neq 1$)

Theoretical Basis

Campbell, Fisher & Justiniano (2017): "Monetary Policy Forward Guidance"

Language conveys **deliberation signals** beyond the rate decision itself — about:

- Future path (forward guidance)
- Risk balance
- Confidence in projections

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- If $\beta = 1$: sentiment is redundant
- If $\beta = 0$: sentiment captures orthogonal info

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H_2: Sentiment explains returns *beyond* surprise

$$R_{i,t} = \alpha_i + \beta_1 S_t + \beta_2 \,\mathrm{Sent}_t + \varepsilon_{i,t} \qquad (2)$$

Test: $\beta_2 \neq 0$ (after controlling for S_t)

Expected ΔR^2

Based on prior literature:

- Bernanke & Kuttner (2005): S_t alone $R^2 \approx 2\%$
- Adding sentiment: expected $\Delta R^2 \approx 1$–3%
- Strongest for equities (language-sensitive)
- Weakest for FX (rates-dominated)

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- Measure: $\Delta R^2 = R^2_{\text{full}} - R^2_{\text{restricted}}$

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Hypothesis 3: Mapping to Structural Shocks

H_3: Sentiment captures the information shock

JK (2020) two-shocks decomposition:

$$\text{Sent}_t = \gamma_0 + \gamma_1 \varepsilon_t^P + \gamma_2 \varepsilon_t^I + u_t \qquad (3)$$

Test: $|\gamma_2| > |\gamma_1|$

Why This Matters

Nakamura & Steinsson (2018):
HFI estimates conflate **two distinct shocks**:

- **Policy:** "Fed lowers rates"
- **Information:** "Fed signals weak economy"

Both compress valuations, but different transmission channels.

Sentiment can **disentangle** them —because language encodes the Fed's *assessment*, not just the decision.

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- If $|\gamma_2| > |\gamma_1|$: sentiment → information, not policy

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Hypothesis 4: Regime-Dependent Effects

H_4: Sentiment effect is stronger during forward guidance

$$R_{i,t} = \alpha_i + \beta_1 S_t + \beta_2 \,\mathrm{Sent}_t + \beta_3 \,(\mathrm{Sent}_t \times \mathrm{FG}_t) + \varepsilon_{i,t} \tag{4}$$

Test: $\beta_3 \neq 0$

Identification Strategy

Three regime periods:

- **Normal** (2000–2008, 2017–2019)
- **ZLB / FG** (2009–2015)
- **Tightening** (2022–2025)

Chow test for structural break; Thompson (2011) double-clustered SE.

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- Expect: $\beta_3 > 0$ for equities, $\beta_3 < 0$ for bonds

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Methodology

Data Sources

Source	Variables	Frequency
FRED	DFF, DGS10, DGS3MO, T10Y3MM, WALCL	Daily
FOMC Statements	Full text (161 statements, 2006–2026)	Event
yfinance	SPY, TLT, GLD, UUP, XLF, XLE, IWM	Intraday
WRDS (CME)	FF futures, Eurodollar futures	Intraday
WRDS (CRSP)	Equity returns, market index; mutual funds (flows)	Daily / Monthly
WRDS (TAQ)	Intraday quotes and trades	Tick-level
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- Event windows: $[-30\text{min}, +30\text{min}]$ and $[-5\text{d}, +5\text{d}]$

Variable Construction: Surprise

Kuttner Surprise (Baseline)

$$S_t = \frac{D}{D-t}\left(F^{\text{FF}}_{t,T} - F^{\text{FF}}_{t-1,T}\right) \tag{5}$$

$F^{\text{FF}}_{t,T}$: FF futures price
D: days in month, t: event day

Bauer-Swanson MPS (Robustness)

$$\text{MPS}_t = \text{PC}_1\left(\Delta \mathbf{r}^{\text{HF}}_t\right) \tag{6}$$

$\Delta \mathbf{r}^{\text{HF}}_t$: vector of 6 intraday rate changes (2y, 5y, 10y, 30y Treasury + FF_{m0}, FF_{m3})

Robustness: results reported under both measures. Bauer-Swanson (2023) shows raw HFI is contaminated by "Fed information effect" —orthogonalization required.

Variable Construction: Sentiment

Hybrid LM + CB Dictionary Score

$$\text{Sent}_t = \frac{1}{2}\left(\frac{N_t^- - N_t^+}{N_t^{\text{total}}} + \frac{H_t - D_t}{N_t^{\text{total}}}\right) \tag{7}$$

- N_t^-, N_t^+: LM negative/positive word counts
- H_t, D_t: CB hawkish/dovish word counts
- Normalized by total words N_t^{total}
- Range: $[-1, +1]$
- > 0: hawkish tone
- < 0: dovish tone

Validation

Cross-validate with **FinBERT** fine-tuned on financial text (Araci, 2019).
Report correlation; use FinBERT as robustness check if $\rho > 0.7$.

Two-Shocks Decomposition (JK Framework)

Sign Restriction (JK 2020)

Asset price response matrix:

$$\begin{pmatrix} \Delta \text{FFR}_t \\ \Delta y_t^{(10y)} \\ \Delta P_t^{\text{eq}} \\ \Delta \text{CS}_t \end{pmatrix} = \mathbf{A} \begin{pmatrix} \varepsilon_t^P \\ \varepsilon_t^I \end{pmatrix}$$

Sign pattern:

$$\text{sign}(\mathbf{A}) = \begin{pmatrix} + & + \\ + & + \\ - & + \\ + & - \end{pmatrix}$$

Why Sentiment Helps Identification

Problem: sign restriction alone cannot distinguish ε^P from ε^I when both compress equity valuations (both give $-$ sign).

Our approach:

1. Run HFI + sign restriction $\rightarrow$ baseline $\hat{\varepsilon}^P, \hat{\varepsilon}^I$
2. Regress Sent_t on both shocks
3. If $|\hat{\gamma}_2| > |\hat{\gamma}_1|$: sentiment captures the **information channel**
4. This provides textual validation of the structural decomposition

Innovation: First study to **cross-validate** sign-restricted shocks with NLP-derived sentiment.

Newey-West HAC (Baseline)

$$\hat{\Omega}^{\text{NW}} = \hat{\Sigma} + \sum_{\ell=1}^{L} \omega_\ell \left(\hat{\Gamma}_\ell + \hat{\Gamma}_\ell' \right)$$

Bartlett kernel: $\omega_\ell = 1 - \frac{\ell}{L+1}$, $L = 4$ (quarterly)

Thompson (2011) Double-Clustered

$$\hat{V} = \hat{V}_{\text{time}} + \hat{V}_{\text{asset}} - \hat{V}_{\text{time} \times \text{asset}}$$

Clusters by FOMC meeting *and* asset class; addresses cross-sectional and time-series dependence simultaneously.

Robustness Checks

- White (1980) heteroskedasticity-robust SE as benchmark
- IV estimation: use surprise component of sentiment (orthogonal to S_t) as instrument
- Placebo: pre-FOMC window $[-60\text{min}, -30\text{min}]$ should show no effect

Expected Findings & Contributions

Expected Findings

Hyp.	Expected Result	Basis
H_1	$\beta \approx 0.4$–0.7 (partial overlap)	LM sentiment weakly correlated with rates
H_2	$\Delta R^2 \approx 1$–3% across assets	BK (2005) baseline $R^2 \approx 2\%$
H_3	$\lvert\hat{\gamma}_2\rvert > \lvert\hat{\gamma}_1\rvert$	Sentiment captures Fed's economic assessment
H_4	$\beta_3 > 0$ for equities (ZLB period)	Language is the tool when rates are fixed

Cross-Asset Hierarchy (Expected)

1. Equities: strongest sentiment effect
2. Corporate bonds (credit spread): moderate
3. FX (USD): moderate (language → risk appetite)
4. Treasuries: weakest (rates-dominated)

Regime Hierarchy (Expected)

1. ZLB (2009–2015): strongest —language is the only tool
2. Tightening (2022–2025): moderate — rapid hikes dominate
3. Normal (pre-2008): weakest —rates do the signaling

Three Contributions

Methodological

- First to integrate HFI (Bauer-Swanson) + NLP sentiment + JK decomposition
- Hybrid LM+CB dictionary validated by FinBERT
- Cross-validation of sign-restricted shocks with textual evidence

Empirical

- First systematic evidence on 2022–2025 tightening cycle
- Cross-asset comparison (equities, bonds, FX, commodities)
- Regime-dependent effects across 3 distinct periods

Policy

- Implications for Fed communication strategy
- Quantify market reaction to language vs. rates
- Inform forward guidance design under different regimes

Target journals: *Journal of Monetary Economics* *Journal of Financial Economics* *Review of Finance*

Preliminary Evidence (Corrected Data, Oct 2026)

Preliminary Evidence I: Prices (Corrected Pipeline)

Data corrections (Oct 2026 audit)

Sample driven by statements + surprises: **161 meetings, 2006–2026** (was 117 minutes-tied, ending 2022); body-only statement texts (2008-12-16 and 2022-01-26 re-downloaded); **same-day** returns in percent (the earlier next-day alignment was mislabelled and inflated the tone–return link).

- H_1 (tone $\sim$ Target): **does not survive** — $\beta_T = +0.08$ per 100 net words ($p = 0.36$; 2006–2022), Wald(Target=Path) $p = 0.59$. Tone is a descriptive measure, not an independent shock series.
- H_2 (incremental power): $\Delta R^2 = +0.16$ pp ($p = 0.64$) — the earlier marginal significance came from the next-day alignment.

The surviving headline: forward guidance

$\beta_{\text{Sent}} = -1.01$ per 100 net words ($p = 0.005$), model $R^2 = 32.8\%$ ($N = 56$) in the FG period (Dec 2008 – Dec 2015), vs. $p = 0.35$ outside.

When the rate instrument is constrained, language becomes the primary transmission channel — H_3's regime prediction is confirmed on corrected data.

Preliminary Evidence II: Flows (WRDS re-run, 2006–2026)

Measurement note

Most CRSP funds report TNA quarterly; flows therefore require a gap rule (reports ≤ 3 months apart) and a $|\text{flow}| \leq 0.5\times\text{TNA}$ cap. Point-in-time fund classification (no look-ahead); the pseudo Bauer–Swanson step is removed. Magnitudes shrink $\sim$40% versus the unfiltered variant; signs and significance are preserved.

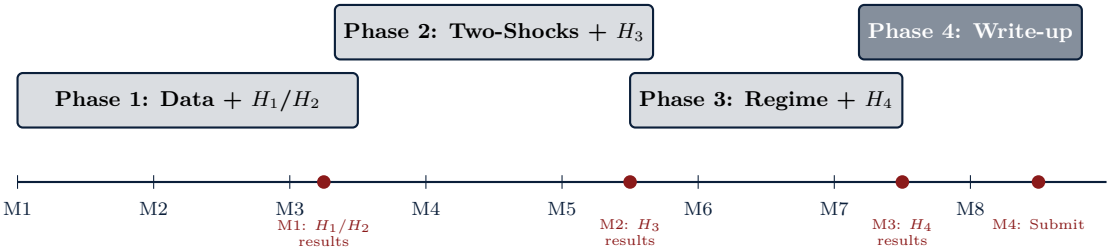
- **Government bonds**: significant inflows on CBI shocks (+0.76% TNA per shock, $p = 0.037$), no MP response; Wald(MP=CBI) $p = 0.098$. The gap-free variant gives +1.13 ($p = 0.010$) — same sign.
- **Risk ladder**: Risk×CBI $= -0.066$ ($p = 0.036$) — CBI shocks push portfolios *down* the ladder (de-risking under information revelations), the **opposite sign** to the original prediction (gap-free: -0.114, $p = 0.002$). Risk×MP flat ($p = 0.88$).

Implication for the hypothesis set

The risk-ladder prediction is revised: central-bank *information* revelations raise perceived uncertainty and trigger de-risking rather than risk-seeking. The sign reversal is reported as found; intraday flow data would discriminate between "prediction wrong" and "monthly aggregation masks faster reallocation".

Research Plan

Timeline


Phase 1: Data + H_1/H_2
Phase 2: Two-Shocks + H_3
Phase 3: Regime + H_4
Phase 4: Write-up
M1
M2
M3
M4
M5
M6
M7
M8
M1: H_1/H_2 results
M2: H_3 results
M3: H_4 results
M4: Submit

Milestones & Deliverables

Phase	Timeline	Deliverable
Phase 1	M1–M2	Data infrastructure complete; H_1, H_2 initial results
Phase 2	M3–M4	Two-shocks decomposition; H_3 results; mid-term paper draft
Phase 3	M5–M6	Regime analysis; H_4 results; robustness checks
Phase 4	M7–M8	Paper finalization; submission to *JME* or *JFE*

Risk Mitigation

- WRDS access delay → fallback to FRED + yfinance
- FinBERT compute → cloud GPU or pre-trained
- Sample size concerns → bootstrap inference

Collaboration

- **Eileen Zhang:** Economic theory, hypothesis development, writing
- **TA / RA:** Data pipeline, NLP implementation, empirical estimation
- **Advisor:** Methodological guidance, submission strategy

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Thank You

Questions & Discussion Welcome

Eileen Zhang | August 2026